

COMMONWEALTH OF AUSTRALIA

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SENG1050

Web Technologies

Week 2 - HTML

Lecture Outline

- ❑ Uniform Resource Identifier (URI)
- ❑ HyperText Markup Language (HTML)
- ❑ Structure of a HTML Document
- ❑ HTML Element Types and Tags

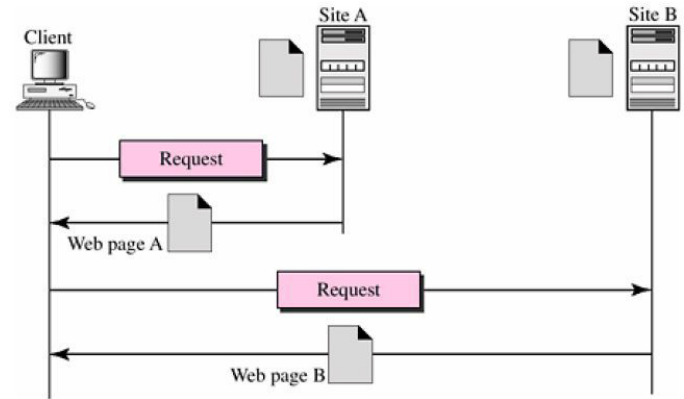


Recall from last week: The Internet

- ☐ We defined our goal by learning about the internet, websites, and how a website can be 'on' the internet.
- ☐ We learned that the internet is a network of networks
- ☐ We learned that websites are accessed using the world wide web, which led us to refine our goal – rather than just creating a website on the internet, it also needs to be accessible using the world wide web.

Recall from last week: The World Wide Web (WWW)

- ❑ The WWW Is a distributed client/server service, in which a client using a browser can access a service using a server.
- ❑ The service provided is distributed over many locations - called sites.
- ❑ Each site holds one or more Web pages.
- ❑ Based on hypertext - instant cross-referencing: links from a word, phrase or image to other pages.
- ❑ A web browser is commonly used to connect to a website's server and display its web pages; but there are also other ways to access this information.



Recall from last week: The World Wide Web (WWW)

Initial idea by Tim Berners-Lee.

☐ <http://www.w3.org/People/Berners-Lee/Overview.html>

Has three aspects:

1. A scheme for locating distributed documents
 - **URI**: URL or URN
2. A protocol for accessing these documents
 - **HTTP** – Covered last week
3. A HyperText language linking the documents together
 - **HTML**

Uniform Resource Identifier (URI)

URI: A scheme for locating distributed documents

The Uniform Resource Identifier (URI) is a string that identifies a resource on the Internet by location, or name, or both.

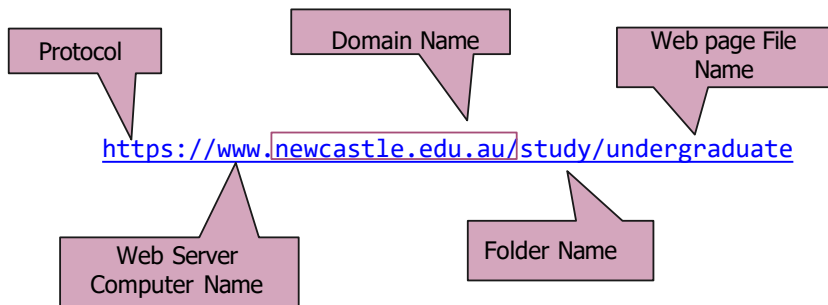
Two specialisations

1. Uniform Resource Locator (URL)
2. Uniform Resource Name (URN)
 - ❑ Does not imply availability of the identified resource
 - ❑ Uses the URN scheme: `urn:<NID>:<NSS>`
 - ❑ Example:
 - `urn:isbn:0451450523`
 - `urn:mpeg:mpeg7:schema:2001`



Uniform Resource Locator (URL)

Represents the network location (physical address) of a resource such as a webpage, a graphic file, or an MP3 file.



Domain Name

A unique name that locates an organisation, a website or other entity on the Internet.

Purpose:

- ❑ Divide the Internet into logical groups
- ❑ Easily memorable names for internet resources
 - Can use <http://www.duckduckgo.com> instead of 20.43.111.112

Organisation: hierarchical

Domain Name

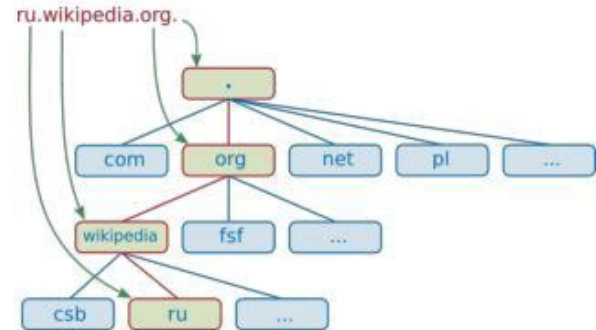
Top-level domain (TLD): identifies the rightmost part of the domain name.

TLD is

- ❑ Either a generic top-level domain: .com, .edu, .org
- ❑ Or a country-code top-level domain: .au, .fr, .np

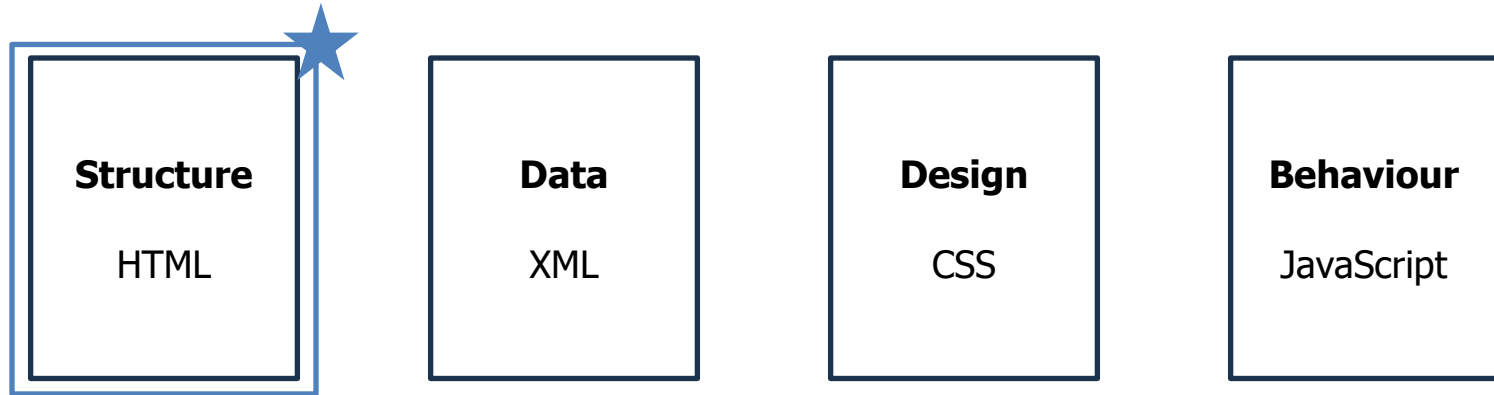
www.newcastle.edu.au

- ❑ Notice that the hierarchical structure is ordered from right-to-left in any given domain name.



HyperText Markup Language (HTML)

Overview of Client-based Web Development





Audiences of your website

Two audiences:

1. Humans
2. Spiders, bots, web-crawlers

Web crawler (e.g., Wikipedia):

- ❑ An Internet bot that systematically browses the WWW, typically for web indexing - to enable quicker searches.
- ❑ Used by search engines to update their web content or indexes or the content of other websites.
- ❑ Can copy all pages they visit for later processing by a search engine

As a web designer, you need to take care of both audiences.

Humans

- ❑ They get the needed information easily and probably in an interesting and nicely presented format.

Spiders, bots, web-crawlers

- ❑ They can identify the information you want to spread and appropriately index them so that:
 - People can find the information they are looking for,
 - Your website has a very high hit count



HyperText

HyperText

"A body of written or pictorial material [emphasis on text] interconnected in a complex way that it could not be conveniently represented on paper. It may contain summaries or maps of its contents and their interrelations; it may contain annotations, additions, and footnotes from scholars who examined it." - Ted Nelson

HyperMedia

The combination of hypertext and multimedia [emphasis on sound, video, etc.] in an electronic document.



HyperText Markup Language (HTML)

- ❑ The file format proposed by Tim Berners-Lee to encode hypertext documents for the WWW
- ❑ Still used for the majority of web pages
 - Although sometimes generated from other sources.
- ❑ HTML structures a web page using tags that:
 - Suggests (not determine) the visual presentation of the document
 - Insert images and other media
 - Add hyperlinks taking the user from this document to another document
 - Create forms that the user can fill in and submit back to the web server
 - Insert small computer programs to be run by the user's browser.

The History of HTML

Version	First draft	Candidate recommendation	Recommendation
HTML 5.0	2007	2012	2014
HTML 5.1	2012	2015	2016
HTML 5.2	2015	2017	2017
HTML 5.3	2017	N/A	N/A

The combined timelines for HTML 5.0, HTML 5.1, HTML 5.2, and HTML 5.3. (from [Wikipedia](#)).

In 2019, HTML5 was replaced entirely with the HTML “Living Standard”, which is the current version. This is maintained primarily by a group of the major browser vendors, the Web Hypertext Application Technology Working Group (WHATWG). For brevity, we will refer to this standard simply as HTML when a version is not specified.

Structure of a HTML Document



HTML Elements

- ❑ A HTML document is a plain text document containing a collection of HTML elements.
- ❑ A HTML element is typically identified/typed using special markup tags.
- ❑ Tags identify different structural elements, such as:
 - ❑ Different levels of headings
 - ❑ Paragraphs
 - ❑ List items, etc.
- ❑ These tags will enclose some document content.
- ❑ A HTML element is a package consisting the tag and content.

```
<element tag> content </element tag>
```

```
<p> this is a paragraph </p>
```



HTML Element Syntax

- ❑ Most HTML elements have **two-sided tags**, i.e., starts with a **start/opening tag** and ends with an **end/closing tag**.
- ❑ Some HTML element do not have any end/closing tag.
- ❑ These elements usually have **empty content (void elements)**.

```
<p> Hi  
There! </p>
```

```
<br>  
<hr>
```



HTML Element Attributes

- ❑ Most HTML elements can have attributes.
- ❑ Attributes **provide additional information** about an element.
- ❑ Attributes are always specified in the start tag.
- ❑ Attributes come in **name/value pairs like: `name="value"`**

Examples:

```
<meta charset="UTF-8">
```

```
<a href="https://www.duckduckgo.com">DuckDuckGo</a>
```



Document Type Definition (DTD)

- ❑ **Purpose:** To identify the type of markup language used in the document.
- ❑ DTD identifies the version of HTML
- ❑ Browser and code validators use that information for processing
- ❑ Placed at the top of the web page document

Common DOCTYPE declarations:

- ❑ HTML5 and Living Standard (current)

```
<!DOCTYPE html>
```

- ❑ HTML 4.01

```
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN"  
"http://www.w3.org/TR/html4/loose.dtd">
```



The Structure of an HTML document

```
<!DOCTYPE HTML>
<html>
  <head>
    <title>Title of the document</title>
    <meta content="text/html; charset=UTF-8" />
  </head>

  <body>
    Content of the document ...
  </body>
</html>
```



The Structure of an HTML document

The opening `<html>` tag marks the start of an HTML document, and the closing `</html>` tag tells a browser that it has reached the end of that HTML document.

`<html>` marks the root element. Anything between these two tags makes up the document content, including all other elements, text, and comments.

```
<!DOCTYPE HTML>
<html>
  <head>
    <title>Title of the document</title>
    <meta charset="UTF-8" />
  </head>
  <body>
    Content of the document ...
  </body>
</html>
```

Inside the `<html>` there must be a single `<head>` and a single `<body>`

`<head>` contains information **about** the document - not (usually) displayed by the browser.

`<body>` contains the **displayed content** of the document.



The Structure of an HTML document

There are several tags that belong inside `<head>`

`<title> ... </title>` is used to define the title of the document.

- ❑ This is not the same as a heading
- ❑ It is often displayed in the browser tab - but it may not be.
- ❑ Provides a title for the page when it is added to favourites.
- ❑ Displays a title for the page in search-engine results.

`<meta>` provides metadata about the HTML document.

- ❑ Metadata will not be displayed on the page, but will be machine parsable.
- ❑ Typically used to specify page description, keywords, author of the document, last modified timestamp, and other metadata.
- ❑ Charset: the character encoding for the HTML document.

```
<!DOCTYPE HTML>
<html>
  <head>
    <title>Title of the document</title>
    <meta charset="UTF-8" />
  </head>

  <body>
    Content of the document ...
  </body>
</html>
```



The Structure of an HTML document

The rest of the tags go inside `<body> ...</body>`

Most HTML elements can be nested

- ❑ One HTML element can contain another HTML element inside it.

```
<!DOCTYPE HTML>
<html>
  <head>
    <title>Title of the document</title>
    <meta charset="UTF-8" />
  </head>
  <body>
    Content of the document ...
  </body>
</html>
```

First HTML document

```
<!DOCTYPE HTML>
<html>
  <head>
    <title>First HTML5 document</title>
    <meta charset="UTF-8" />
  </head>

  <body>
    Hello World!
  </body>
</html>
```

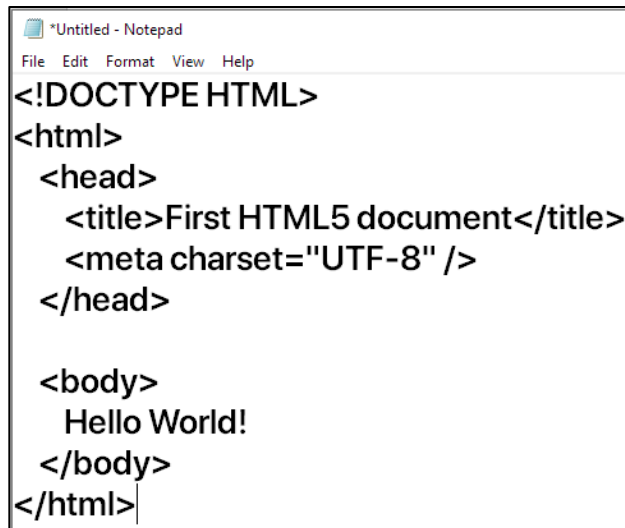
Have a laptop? Try it out now!

Open a text editor (such as Notepad) and copy the code on the left.

Save the file as "helloworld.html" (Under 'Save as Type', select 'All Files' to prevent the automatic .txt extension from being used).

Open the file in your favourite browser (Edge, Firefox, Chrome, Safari, Opera, etc).

First HTML document

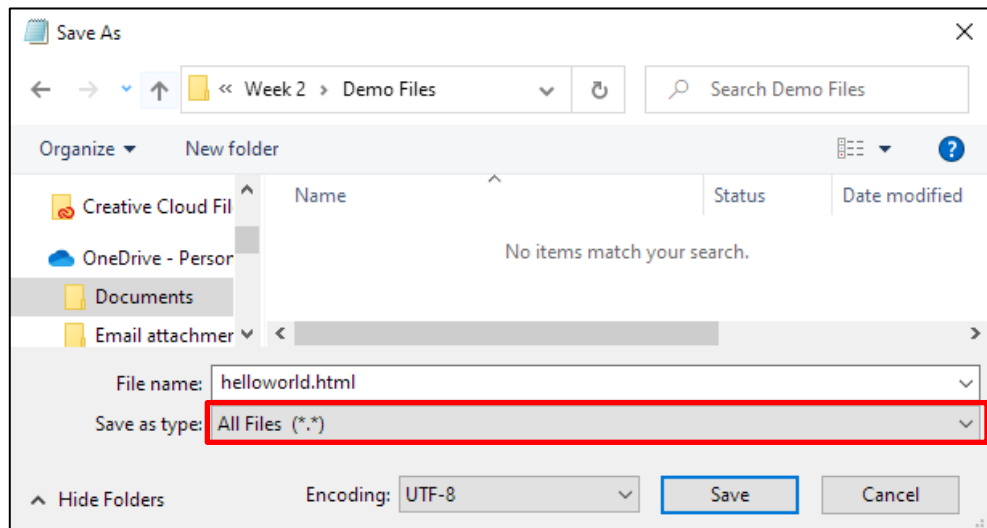


*Untitled - Notepad

File Edit Format View Help

```
<!DOCTYPE HTML>
<html>
  <head>
    <title>First HTML5 document</title>
    <meta charset="UTF-8" />
  </head>

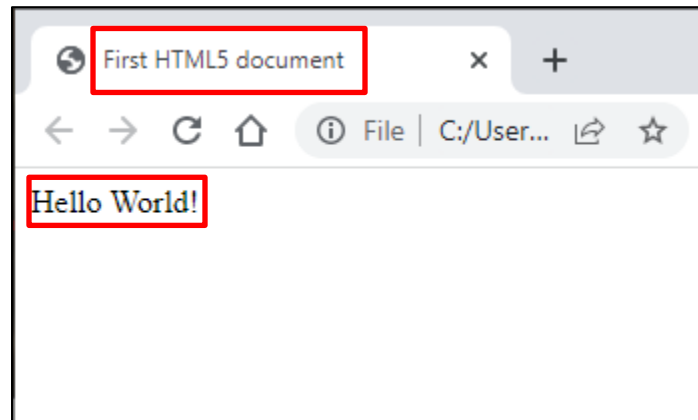
  <body>
    Hello World!
  </body>
</html>
```



First HTML document

```
<!DOCTYPE HTML>
<html>
  <head>
    <title>First HTML5 document</title>
    <meta charset="UTF-8" />
  </head>

  <body>
    Hello World!
  </body>
</html>
```

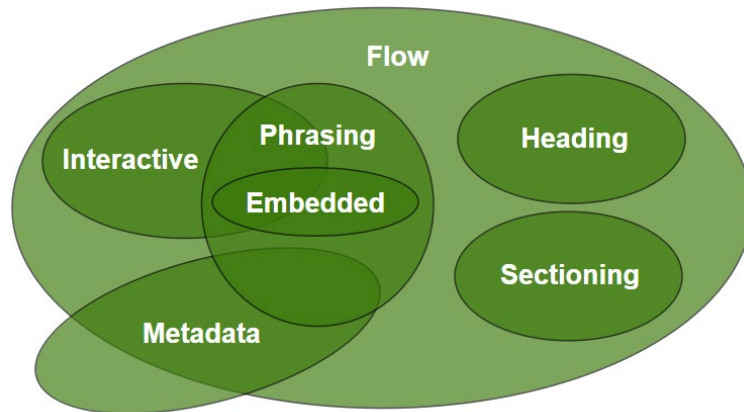


HTML Element Types and Tags

HTML Element Types

HTML elements are grouped into several types, some of which overlap:

- ☐ Flow content
- ☐ Heading content
- ☐ Phrasing content
- ☐ Embedded content
- ☐ Interactive content
- ☐ Metadata content
- ☐ Sectioning content



Source: <https://html.spec.whatwg.org/>



h1, h2, h3, h4, h5, h6 Tags

- ❑ HTML headings are titles or subtitles that you want to display on a webpage.
- ❑ Defined with the `<h1>` to `<h6>` tags, with `<h1>` being the most important and `<h6>` being the least important.
- ❑ **Headings are important:**
 - Search engines use headings to index the structure and content of the webpage, as well as rank documents.
 - Headings are important to show the document structure to the users.
 - But don't spam your document with too many headings. Google doesn't like that.
 - Use HTML heading tags for headings only, not for formatting purposes.

```
<h1>h1 tag</h1>
<h2>h2 tag</h2>
<h3>h3 tag</h3>
<h4>h4 tag</h4>
<h5>h5 tag</h5>
<h6>h6 tag</h6>
```



p, blockquote Tags

<p> ... </p>

- ❑ A paragraph of text. No special default formatting
- Be careful – If you forget </p>, the start of the next block-level element will automatically imply it. It is better to explicitly specify this closing tag.

```
<body>
  <h1>About WWF</h1>
  <p>Here is a quote from WWF's website</p>
  <blockquote cite="http://www.worldwildlife.org/who/index.html">
    For 50 years, WWF has been protecting the future of nature.
  </blockquote>
</body>
```

<blockquote> ... </blockquote>

- ❑ A block of text quoted from someone/somewhere else.
- ❑ Formatted as an indented block of text by default.
- ❑ Attribute `cite`: value => source of the quotation.
- ❑ Don't use blockquote just to indent the text.

About WWF

Here is a quote from WWF's website

For 50 years, WWF has been protecting the future of nature.

address Tag

`<address> ... </address>`

- ❑ Contact information for the author/owner of a document or an article
- ❑ Usually rendered in *italics* by default. Most browsers will also add a line break before and after the address element.
- ❑ Inside the **<body>** element: contact information for the document
- ❑ Inside an **<article>** element: contact information for that article.

```
<body>
  <h1>Contact details</h1>
  <address>
    Mickey Mouse <br>
    Room MMXYZ <br>
    Disneyland, USA <br>
  </address>
</body>
```

Contact details

*Mickey Mouse
Room MMXYZ
Disneyland, USA*



div Tag

`<div>...</div>`

- ❑ A generic division of the document.
- ❑ Defines a division or a section in an HTML document
- ❑ No special formatting by default.
- ❑ Used to group tags together and apply formatting.
- ❑ Used to group other flow content elements to format them with CSS.



br, hr Tags

Void elements refer to elements that do not have a closing tag and so contain no content. For example:

- ❑ Goes to the next line before displaying the next element or portion of the text.
- ❑ Not a paired element; it is a standalone or void element.
- ❑ Use the
 tag to enter line breaks, not to separate paragraphs.

<hr>

- ❑ Draws a horizontal rule between blocks to visually separate those areas.
- ❑ Also a void element.

With the latest HTML, don't use
 or <hr />

Flow Content Example

```
<body>
  <p>This is some text before the article.</p>
  <hr>
  <div>
    <h2>This is the heading in a div element</h2>
    <p>This is content of the article in a div element.</p>

    <address>
      Written by: Mickey Mouse.<br>
      Visit us at:<br>
      Example.com<br>
      Box 564, Disneyland<br>
      USA
    </address>
  </div>
  <hr>
  <p>This is some text after the article.</p>
</body>
```

This is some text before the article.

This is the heading in a div element

This is content of the article in a div element.

Written by: Mickey Mouse.

Visit us at:

Example.com

Box 564, Disneyland

USA

This is some text after the article.

Phrasing Content

Phrasing content refers to HTML elements that primarily markup the content within paragraphs and individual phrases of the document.

- ❑ Usually displayed by browsers as inline elements (that is, an element that is displayed without starting a newline.
- ❑ For example: hyperlinks, emphases, images
- ❑ Most phrasing content cannot contain other HTML elements that are not also phrasing content.
- ❑ For example, the following HTML will not validate:

```
<span><p>Some text</p></span>
```

This is invalid because `<span>` is phrasing content, but `<p>` is not.



strong, em, mark, u, b, i Tags

Inline tags are placed around phrases of text

- ❑ To impart a semantic meaning (e.g., importance or emphasis)
- ❑ To format the appearance

`<strong>...</strong>`

- ❑ The enclosed text is important
- ❑ Displayed in **bold** by default.

`<em>...</em>`

- ❑ The enclosed text should be emphasised.
- ❑ Displayed *italicised* by default.

`<mark>...</mark>`

- ❑ Indicates text that is of particular relevance to the user (such as search terms). Compare with **strong**, used for text that is important.
- ❑ Displayed **highlighted** by default.

`<u>...</u>`

- ❑ The enclosed text has a non-textual annotation (such as indicating a spelling error).
- ❑ Displayed underlined by default.

- ❑ `<b>...</b>` makes the enclosed text **bold**, whereas `<i>...</i>` makes it *italics*. Semantically, `<b>` means 'bring attention to', whereas `<i>` indicates idiomatic text (such as technical terms, or an alternative voice in a sentence).
- ❑ These tags might look the same as `<strong>` and `<em>` to humans, but they are not the same to spiders.
- ❑ Use `<b>` and `<i>` as a last resort when other options are not appropriate.

Recommendations:

Headings	<code><h1></code> to <code><h6></code>
Emphasise text	<code></code>
Important text	<code></code>
Marked/highlighted text	<code><mark></code>
Definition term	<code><dfn></code>
Title of a cited work	<code><cite></code>

Phrasing Content Example

```
<body>
  <p>
    I want to <em>emphasise</em> this but <i>not this</i>.<br><br>
    In fact, I want to <strong>strongly emphasise</strong> this but <b>not this</b>.
  </p>
</body>
```

I want to *emphasise* this but *not this*.

In fact, I want to **strongly emphasise** this but **not this**.

code, samp, kbd, var Tags

<code><code>...</code></code>	Piece of computer code
<code><samp>...</samp></code>	Sample of (program) output
<code><kbd>...</kbd></code>	Input that would be typed on a keyboard
<code><var>...</var></code>	The enclosed text is a variable in a programming context

- ❑ A set of elements that apply a specific type of text content.
- ❑ Initially existed out of a need for formatting but have since come to assume a new role as semantical text identifiers in HTML5.
 - To add information about the type of text contained in the tag.

```
<p>  
  If you compile the C code  
  <code>scanf("%f", &scale);  
    printf("%f\n", scale*2);  
  </code>  
  and enter <kbd>21.12</kbd> on  
  the keyboard, then the program  
  will output <samp>42.24</samp>.  
</p>
```



If you compile the C code `scanf("%f", &scale);
printf("%f\n", scale*2);` and enter 21.12 on the
keyboard, then the program will output 42.24.

cite, q Tags

<cite>...</cite>

- ❑ Indicates that the enclosed text is the **title** of a work.
- ❑ Often used in conjunction with a quotation or as a reference.
- ❑ **Note:** Prior to HTML5, <cite> tag defined a definition.

<q cite=" " >...</q>

- ❑ The enclosed text is a **short quotation** from another source.
- ❑ The cite attribute provides the URI of the source.
- ❑ This is a phrasing element for showing quotations.
Compare to <blockquote>

```
<p>
  There is a computing "urban legend" that Bill Gates once said,
  <q cite="http://archive.wired.com/politics/law/news/1997/01/1484">
    640K of memory should be enough for anybody
  </q>
  (Jon Katz, <cite>Did Gates Really Say 640K is Enough for Anyone?</cite>,
  Wired News, 16 Jan 1997).
</p>
```

There is a computing "urban legend" that Bill Gates once said, "640K of memory should be enough for anybody" (Jon Katz, *Did Gates Really Say 640K is Enough for Anyone?*, Wired News, 16 Jan 1997).

dfn, abbr Tags

<dfn>...</dfn>

- ❑ The enclosed text is a **definition term**.
- ❑ The nearest tag to the **<dfn>** tag must contain the definition/explanation.

```
<p>  
  <dfn>HTML</dfn> is the standard markup  
  language for creating webpages.  
</p>
```

HTML is the standard markup language for creating webpages.

<abbr>...</abbr>

- ❑ Indicates that the enclosed text is an **abbreviation or an acronym**
- ❑ By marking up abbreviations appropriately, you can provide useful information to browsers, spell checkers, translation systems, and search engine indexers.

```
<body>  
  <p>The <abbr title='World Health  
  Organisation'>WHO </abbr> declared  
  a pandemic.</p>  
</body>
```

The WHO declared a pandemic.

World Health
Organisation

sup, sub Tags

^{...}

- The enclosed text is a mathematical **superscript**.
- Example: ABC^{xyz}

_{...}

- The enclosed text is a mathematical **subscript**
- Example: H_2O

```
<body>
  <p>
    <h2>Superscript</h2>
    ABC<sup>xyz</sup>

    <h2>Subscript</h2>
    H<sub>2</sub>O
  </p>
</body>
```

Superscript

ABC^{xyz}

Subscript

H_2O



span Tag

`<span>...</span>`

- ❑ Generic grouping of inline text - used to apply formatting.
- ❑ Does the same job as `<div>` but does not start a new block of text.
- ❑ Provides no visual change by itself.
- ❑ Provides a way to add a hook to a part of a text or document.
 - When text is hooked in a `<span>` element, you can style it with CSS, or manipulate with JavaScript.



ol, ul, li Tags

Special block tags let you create lists.

`<ol>...</ol>`

An **ordered list** of items

Attributes:

- ❑ **start**=number; specifies the start value of an ordered list
- ❑ **type**={"1,A,a,I,i"}; specifies the kind of marker used in the list
- ❑ **reversed**; specifies that the list order should be descending.

`<ul>...</ul>`

An **unordered** list of items

`<li>...</li>`

- ❑ An **individual item in a list**.
- ❑ Used in both `<ul>` and `<ol>`.

Attribute:

- ❑ **value**=number; specifies the value of a list item. Each following list item will increment from that number (only for `<ol>`)

ol, ul, li Tags Example

```
<body>
  <h1>Taking a Good Photo</h1>
  <ol>
    <li>Find a nice place.
      <ul>
        <li>Taree is nice.</li>
        <li>Aliens in Taree are nicer.</li>
      </ul>
    </li>
    <li>Make sure you have a camera.
      <ul>
        <li>If you don't, borrow one.</li>
        <li>You can even buy one from a camera shop.</li>
      </ul>
    </li>
    <li>Take off the lens cap.</li>
    <li>Etc.</li>
  </ol>
</body>
```

Taking a Good Photo

1. Find a nice place.
 - Taree is nice.
 - Aliens in Taree are nicer.
2. Make sure you have a camera.
 - If you don't, borrow one.
 - You can even buy one from a camera shop.
3. Take off the lens cap.
4. Etc.

dl, dt, dd Tags

`<dl>...</dl>`

- ❑ A description list
- ❑ Used in conjunction with `<dt>` and `<dd>`

`<dt>...</dt>`

- ❑ Used within a description list to indicate a term/name to be defined

`<dd>...</dd>`

- ❑ Used within a description list to describe a term

- ❑ List tags should only contain the appropriate item tags as immediate children.

- ❑ `<dt>` cannot contain heading or sectioning content.

- ❑ `<li>` and `<dd>` can contain inline and block tags - allows nested lists!

dl, dt, dd Tags Example

```
<body>
  <dl>
    <dt>HTML</dt>
    <dd>HyperText Markup Language</dd>

    <dt>SMTP</dt>
    <dd>Simple Mail Transfer Protocol</dd>

    <dt>FTP</dt>
    <dd>File Transfer Protocol</dd>
  </dl>
</body>
```

HTML

HyperText Markup Language

SMTP

Simple Mail Transfer Protocol

FTP

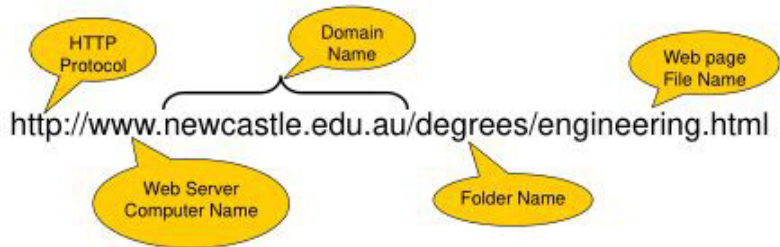
File Transfer Protocol



a Tag

`<a href="uri">...</a>`

- ❑ Create a hyperlink from the enclosed content to the document named in the URI
- ❑ Called an **anchor**
- ❑ When the link is activated in the browser, the linked document is requested and (all going well) displayed in the browser.
- ❑ `<a>` is an interactive content element.





Absolute URL vs. Relative URL

Absolute URL

- ❑ Enough information to access the document from any other document
- ❑ At the very least, protocol, host and path are specified.

Relative URL

- ❑ Only enough information if the context of the current document is used.

Relative URL

- ❑ Protocol/host default to same as current document
- ❑ Port defaults to number appropriate for protocol (http = 80, ftp = 21); host can override this
- ❑ Path defaults to the same as the current document
- ❑ Paths without leading / are relative to current path
- ❑ Name default determined by host (usually index.html)
- ❑ Id defaults to nothing (top of document)
- ❑ Query defaults to nothing

Absolute URL vs. Relative URL Example

- ❑ I am currently adding a link to (editing the file)
<http://compseng.newcastle.edu.au/abc123/index.html>
- ❑ I want to put in a link to the research webpage
<http://compseng.newcastle.edu.au/abc123/research.html>
- ❑ I could use the absolute URL
`<a href="http://compseng.newcastle.edu.au/abc123/research.html">Current Research</a>`
- ❑ Better yet, I could use the relative URL
`<a href="research.html">Current Research</a>`

Absolute URL vs. Relative URL Example

- ❑ I am editing the research html file at
<http://compseng.newcastle.edu.au/abc123/lab/research.html>
- ❑ I want to link to the main page at
<http://compseng.newcastle.edu.au/abc123/index.html>
- ❑ I could use the relative URL
`<a href="../index.html">Back to the main page</a>`
- ❑ However, since index.html is the default name on the compseng.newcastle.edu.au web server, I can use
`<a href="..">Back to the main page</a>`

Absolute URL vs. Relative URL Recommendations

- ❑ If the document you are linking to is part of your website, then **use relative URLs**.
- ❑ Only if the document is part of a different website (on a different server, or on the same server but controlled by someone else) should you use absolute URLs.
- ❑ **Important:** This way, you can 'zip' the entire website, move it to another website, and it will still work.

To check whether your links are correct (correct URL structure and point to existing documents), you can:

- Manually try every link to see if it takes you to the correct document, or
- Use an automatic link checking tool, such as <http://validator.w3.org/checklink>



a Tag

Linking to Identifiers

- ❑ The URL `/jfr457/research/index.html#gd` requires there to be something in the requested document that says “identifier gd points to this spot”.
- ❑ For any HTML tag: `<tag id="gd">...</tag>` (for a two-sided tag) or `<tag id="gd">` (for an empty tag).
- ❑ Every tag (almost) allows the id attribute. Note that an id must be unique within the document.
- ❑ If there is no matching id attribute, then the browser will point to the bottom of the document, or the top, or where you were the last time you loaded this document
 - Which can cause unnecessary confusion.
- ❑ For example:

```
<h2 id="about">About Us</h2>
<a href="#about">Go to About Us!</a>
```



img Tag

```

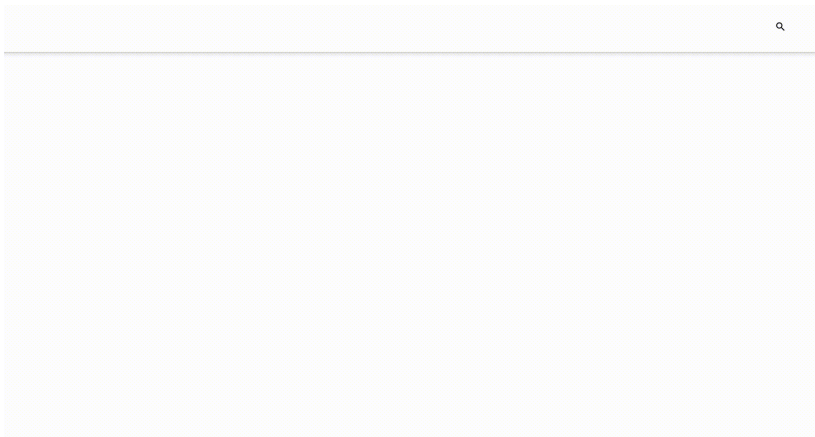
```

- ☐ Loads the image file from uri and inserts it into the document at this point.
- ☐ The image is displayed with 42 pixel height x 42 pixel width.
- ☐ If the image cannot be loaded or displayed, then displays the alternative text instead.
 - ☐ Error messages are a common use for this, for example, alt="Image could not be displayed"
- ☐ is a phrasing content element.
- ☐ Src and alt are required attributes.
- ☐ WebP, JPEG, GIF and PNG image formats are almost universally supported; avoid system-specific formats like BMP, XBM, PICT

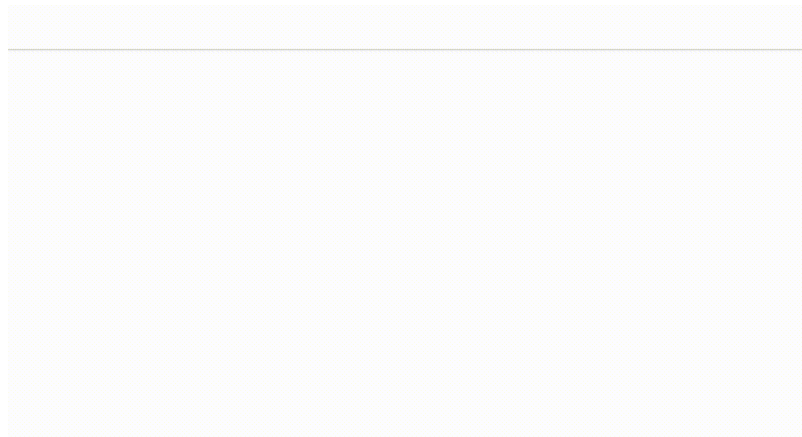
HTML Images

- ☐ Images are one of the biggest sources of slow webpage loading
 - ☐ Use WebP where possible with compression for the smallest possible images.
 - ☐ Avoid images greater than 1 MB unless necessary.
- ☐ Always set both the width and height attributes (this will cause the browser to reserve the required space during page load so the layout doesn't suddenly change when the image loads).

HTML Images



Bad: Width and Height attributes not set.
Notice how the text jumps down when
the images load.



Good: Width and Height attributes are set. The
browser reserves space so the layout does not
change when the images load.

HTML Special Characters

HTML can display the full range of Unicode characters independent of the `<?xml?>` encoding

`&#number;`

Displays character with given Unicode number

`&name;`

Displays a “named” character

Some common characters	
<code>&amp;</code>	& (ampersand)
<code>&lt;</code>	< (less than)
<code>&gt;</code>	> (greater than)
<code>&eacute;</code>	é (e with acute accent)

You must use `&`, `<`, `>`; NOT `&` `<` `>`



HTML Comments

`<!-- comment text -->`

- ☐ Any text between `<!--` and `-->` is ignored by the browser.
- ☐ Use this to embed notes for yourself and other web developers in your web pages.
- ☐ Just like any other programming language, you should add a comment when the author's intent is not immediately clear from the code.
- ☐ Commenting can also be used to embed metadata, such as authorship and keywords - but there are other ways to do this ...

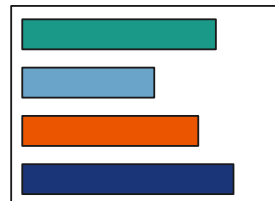


HTML Element Display

There are multiple ways for browsers to render a given HTML element, the two most common types are block display and inline display:

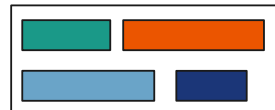
Block Display

- ❑ Elements displayed in the block format normally start (and end) with a new line when displayed in a browser.
- ❑ Most HTML elements (except phrasing content) are displayed in this way
- ❑ For example: paragraphs, headings, lists, tables



Inline Display

- ❑ Elements displayed in the inline format are normally displayed without starting a new line.
- ❑ Most phrasing content is displayed in this way.
- ❑ For example: hyperlinks, emphases, inline images





HTML Do Not Define Visual Formatting

Almost all tags imply suggestions to the browser about how to draw their content.

- ❑ Headings change the font size and weight
- ❑ `<blockquote>` indents the margins

But, these are only suggestions!

- ❑ A browser can ignore them
- ❑ It may not even support them



HTML Validation

If there is a mistake in your HTML

- ❑ Some browsers will refuse to display the page.
- ❑ Some browsers will report that the page is invalid but try to display it anyway.
- ❑ Some browsers will try to guess what it thinks you want to display.

Just because it looks right on your browser,
doesn't mean it is right!

For all the gory details, consult:

- ❑ <http://www.w3.org/TR/html5>
- ❑ <http://www.w3.org/TR/xhtml1>
- ❑ <http://www.w3.org/TR/html4>

Learn to use an XML/HTML validator

- ❑ Checks your pages to ensure they conform to the standards
- ❑ <http://validator.w3.org/>
- ❑ <http://www.htmlhelp.com/tools/validator/upload.html>

All HTML you write for your assignments MUST validate – huge mark loss if validation fails, even if the page still displays.

Which of the following parts of a webpage are defined by HTML tags? You can choose up to TWO answers.

The structure of the page.

0%

The semantic meaning of each element.

0%

The way each element should be displayed.

0%

The behavior of the page when the user interacts with each element.

0%



Which tag will a search engine use to determine the title of a webpage in search results?

meta

0%

h1

0%

heading

0%

header

0%

title

0%



Which HTML tag would be semantically correct to use for displaying unimportant, irrelevant text in bold?

strong

0%

em

0%

b

0%

i

0%

u

0%

mark

0%

None of the above

0%



Which of the following tags can be direct children of an 'ol' or 'ul' element?

p

0%

h2

0%

blockquote

0%

li

0%

dd

0%



```
<!DOCTYPE html>
<html lang="en">
  <head>
    <title>Spot the error</title>
    <meta charset="UTF-8">
  </head>

  <body>
    <p>This <dfn>HTML</dfn> document has an <u>errorrr</u>. No, not that spelling error.</p>
    <p>Can you spot it? <i>Hint: you're looking for a tag being used in the wrong semantic context...</i></p>
  </body>
</html>
```



Which tag contains error?

meta

0%

p

0%

dfn

0%

u

0%

i

0%



Which of the following is a relative path that could be used by a HTML file stored at "D:/webroot/html/www/mypage.html" to link to an image located at "D:/webroot/images/background.png"?

D:/webroot/images/background.png

0%

/images/background.png

0%

background.png

0%

../../images/background.png

0%

webroot/../images/background.png

0%

D:/webroot/../../background.png

0%



Which of the following demonstrates the correct way to include an image in an HTML file?
Select ALL the correct answer.

A

`` 0%

B

`` 0%

C

``

D

``



Which HTML tag is used for making text appear bold?

strong

0%

b

0%

mark

0%

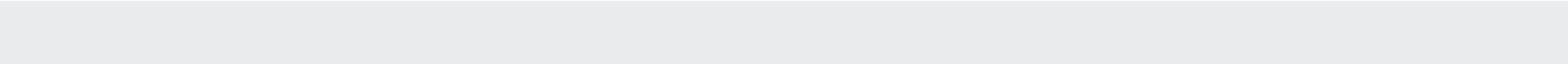
em

0%

None of the above

0%





Summary

- ❑ We have learned the three elements of the world wide web, the scheme for locating documents (URI), the protocol for accessing them (HTTP), and the language to write them in (HTML).
- ❑ We can write HTML documents to define the structure of our webpages and display data on them.
- ❑ But this reveals a new problem: HTML mixes both the structure of the webpage and the data together. If the data ever changes, we would have to update every single HTML file using that data, a big problem on larger websites!
- ❑ To address this, we should separate the structure of our website from the data. For this, we need XML – next week.

Summary

- ❑ What is HTML?
- ❑ Basic HTML elements
- ❑ Types and attributes of HTML elements
- ❑ Difference between block and inline display
- ❑ Importance of choosing proper tags so that spiders can identify the importance of your page
- ❑ Phrasing content elements
- ❑ Different types of list elements
- ❑ How to link different HTML documents

References

- ❏ Web Development and Design Foundations with HTML5 (7e, 9e)
 - By Terry Felke-Morris
 - Chapters 1 & 2
- ❏ <http://www.w3schools.com/html/default.asp>
- ❏ <http://www.w3.org> - the first thing to look for all things related to Web standards